

Introduction to General Agent Problem Solving

AI-101: Moving from Code to Autonomous Agents

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COURSE LEARNING OBJECTIVES

Architectural Shift

Transition from a standard software developer to a **Certified Agentic AI Architect**. Learn to build "Digital FTEs."

Core Frameworks

Master **Spec-Driven Development (SDD)** and the 10-80-10 operating model for production-ready agents.

The Mindset Shift

"No Postponing" & The Startup Mentality

AUTOMATION VS. AUTONOMY

The Smart Microwave Analogy

- **Traditional:** Human judges food weight, moisture, and manually sets time/power.
- **Autonomous:** System uses **sensors** (start temp, steam) to decide its own run-time.
- The transition is from *following instructions* to *solving problems* using environment perception.



THE TWO PILLARS OF AGENTIC AI

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Core Components

1. Perception (Sensors)

Agents must read parameters dynamically from their environment (logs, user intent, data files).

2. Action (Effectors)

Agents take actions through effectors to change the environmental state and achieve goals.

THE GAPS FRAMEWORK OVERVIEW



Goal Formulation

Defining exactly what "done" looks like and setting the success criteria.



Environment Search

Exploring the state space to find the optimal path to the goal.



Execution

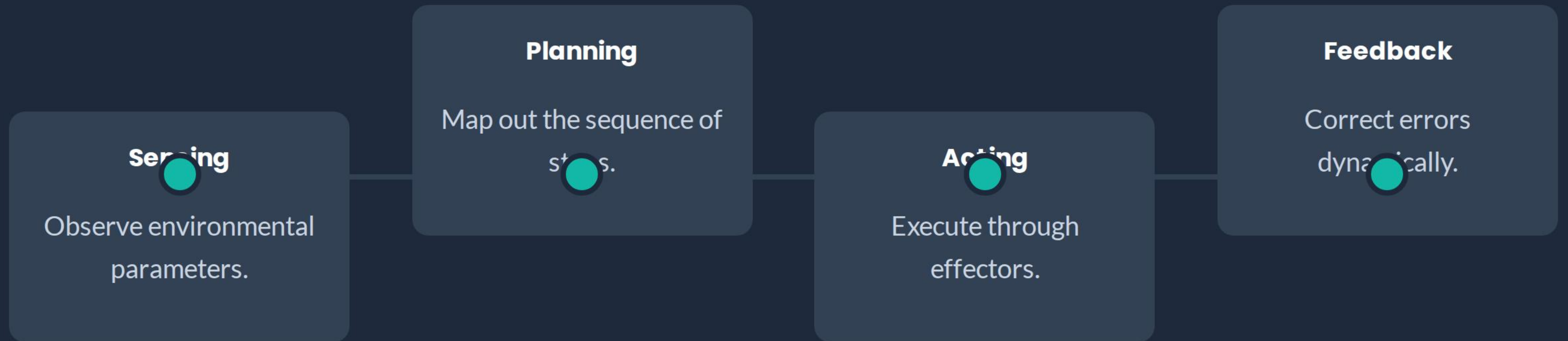
Deploying actions via effectors and monitoring feedback loops.

THE 10-80-10 OPERATING MODEL



- 10% **Direct**: Set Intent & Specs
- 80% **Execute**: AI Agent Reasoning
- 10% **Verify**: Human Review & Quality

THE AGENTIC EXECUTION LOOP

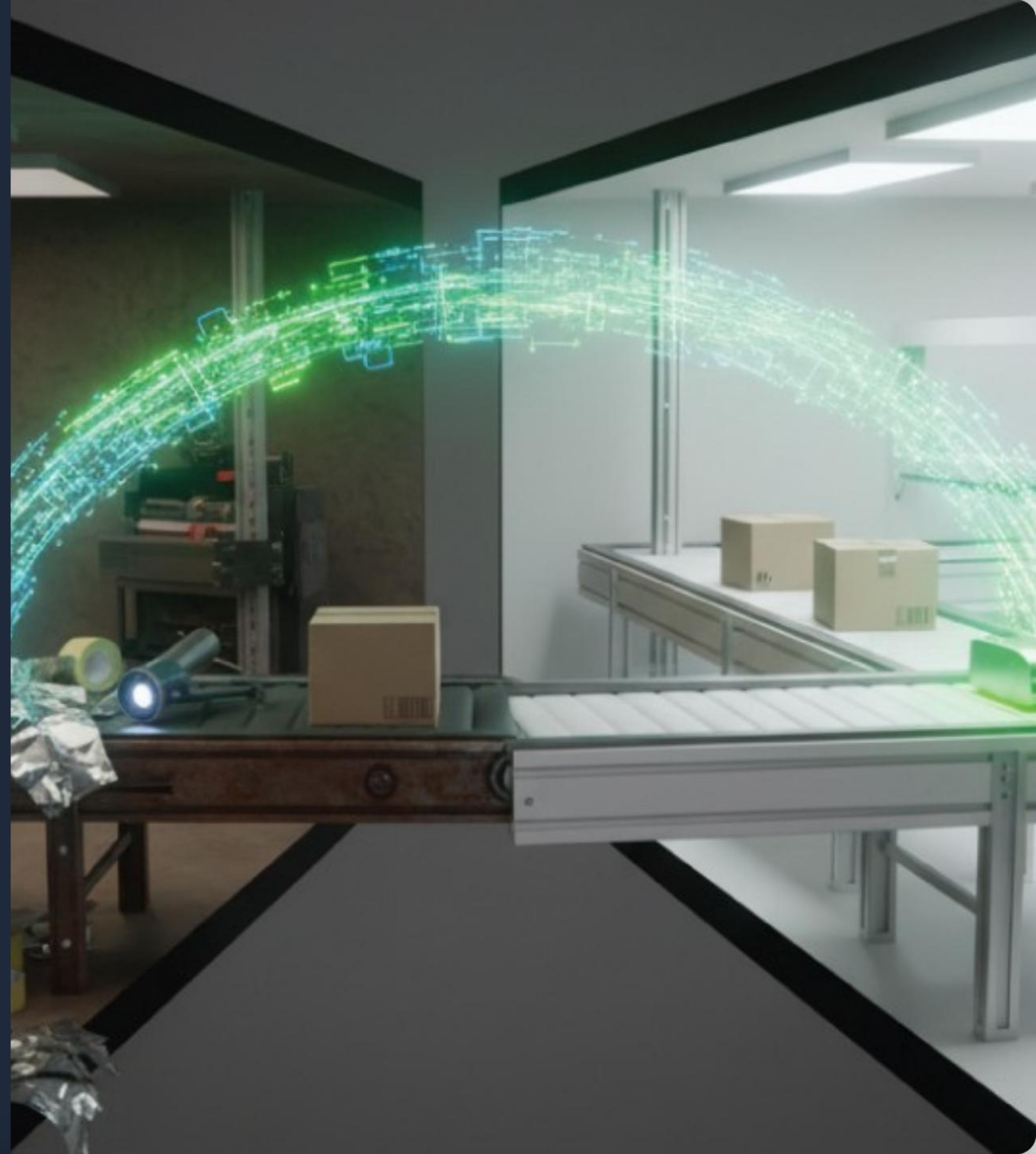


SAFETY & SECURITY

Human-in-the-Loop (HITL)

Agents are restricted from high-risk "Write Access" without explicit human authorization. This mitigates the risk of unpredictable autonomous actions.

Guardrails: The agent builds the draft; the professional judges and approves the execution.



INCUBATORS VS. SPECIALISTS

Feature	General Agent (Incubator)	Custom Agent (Specialist)
Purpose	Explore, discover, prototype	Execute, scale, govern
Reliability	Variable (creative/exploratory)	Consistent (deterministic)
Governance	Human-in-the-Loop focus	Guardrails built by design
Cost Profile	Higher per-request	Optimized for high volume

THE PATH TO MASTERY

-  **Literature Review:** Deeply analyze foundational thesis papers on problem-solving agents.
-  **Hands-on:** Move immediately from conceptual learning to building your first agent.
-  **Community:** Share patterns, edge cases, and "Digital FTE" recipes with the cohort.



Any Questions?

Panaversity: Agentic AI Architect Certification

Next Step: Your First Agent Build

www.panaversity.org

IMAGE SOURCES



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